



HACKTHEBOX



Pursue the Tracks

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Difficulty: Easy

Classification: Official

Synopsis

- Pursue The Tracks is an easy forensics challenge involving extracting information from MFT records and answering questions to complete the investigation and get the flag.

Description

- Luxx, leader of The Phreaks, immerses himself in the depths of his computer, tirelessly pursuing the secrets of a file he obtained accessing an opposing faction member's workstation. With unwavering determination, he scours through data, putting together fragments of information trying to take some advantage on other factions. To get the flag, you need to answer the questions from the docker instance.

Skills Required

- Basic windows knowledge

Skills Learned

- extracting information from MFT
- Analyzing an incident based on MFT records

Enumeration

Players are provided with the following file:

- z.mft

The provided file is only the MFT component of a virtual disk.
Because of this, the user can parse it with the appropriate tools.

A well-known tool is [MFTECmd](#)

```
MFTECmd.exe -f z.mft --csv c:\temp
```

It will export the entire content to a CSV file. This file can be opened with [TimelineExplorer](#), [Excel](#), or any other program of your choice. Of course, it would be better if one convert it to [xls](#) or imports the data to Google Sheets for easier filtering.

A different way to parse the file is to directly open it with [MFTExplorer](#) or [MFT_Browser](#). However, some tools might miss important information such as the timestamp or file size.

For this challenge, the first approach is better because we need to reconstruct the timeline of the events.

Solution

Note: do not count system files while replying to questions.

1. Files are related to two years, which are those? (for example: 1993,1995)

Filtering for "" + "FileName" columns:

	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	nUse	Parent	Parent	Parent	FileName	Extension	FileSize	Refere	Repars	IsDirectory	HasAds	IsAds	SI<FN	uSecZel	Copied	SiFlags
7	TRUE	5	5	.							FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
16	TRUE	5	5	.\$Extend							FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
20	TRUE	11	11	.\$Extend	\$RmMetadata						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
23	TRUE	11	11	.\$Extend	\$Deleted						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
24	TRUE	27	1	.\$Extend	\$Txflg						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
25	TRUE	27	1	.\$Extend	\$Txf						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
29	TRUE	5	5	.	System Volume Information						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
31	TRUE	5	5	.	documents						FALSE	FALSE	FALSE	FALSE	FALSE	None
32	TRUE	36	1	.\docume		2023					FALSE	FALSE	FALSE	FALSE	FALSE	None
39	TRUE	36	1	.\docume		2024					FALSE	FALSE	FALSE	FALSE	FALSE	None

Answer: 2023, 2024

2. There are some documents, which is the name of the first file written? (for example: randomname.pdf)

Entries are incremental, so the first file written is the first one with the column "IsDirectory = False".

InUse	Parent1	Parent2	Parent3	FileName	Extension	FileSize	RefersTo	Repars	IsDirectory	HasAds	IsAds	Si<FN	uSecZe	Copied	SiFlags
TRUE		5	5	\$MFT		26				FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$MFTMirr		209				FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$LogFile						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$Volume						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$AttrDef						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$Bitmap						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$Boot						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$BadClus						TRUE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$BadClus:\$Bad		936				FALSE	TRUE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$Secure		26				TRUE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$Secure:\$SDS		26				FALSE	TRUE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$UpCase		13				TRUE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE		5	5	\$UpCase:\$Info						FALSE	TRUE	FALSE	FALSE	FALSE	Hidden
TRUE	11	11	11	.\$Extend \$Quota						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE	11	11	11	.\$Extend \$ObjId						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE	11	11	11	.\$Extend \$Repair						FALSE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE	27	1	1	.\$Extend \$Repair						TRUE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE	27	1	1	.\$Extend \$Repair:\$Config						FALSE	TRUE	FALSE	FALSE	FALSE	Hidden
TRUE	30	1	1	.\$Extend \$Tops						TRUE	FALSE	FALSE	FALSE	FALSE	Hidden
TRUE	30	1	1	.\$Extend \$Tops:\$T		104				FALSE	TRUE	FALSE	FALSE	FALSE	Hidden
TRUE	30	1	1	.\$Extend \$TxLog.blf	.blf	6				FALSE	FALSE	FALSE	FALSE	FALSE	Archive
TRUE	34	1	1	System\WPSettings.dat	.dat					FALSE	FALSE	FALSE	FALSE	FALSE	Archive
TRUE	37	1	1	.\documeFinal_Annual_Report.xlsx	.xlsx	3				FALSE	FALSE	FALSE	FALSE	FALSE	Archive
TRUE	37	1	1	.\documeFinal_Financial_Statement.xlsx	.xlsx	59392	1		FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive
TRUE	37	1	1	.\documeFinal_Project_Proposal.pdf	.pdf	57344	1		FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive
TRUE	37	1	1	.\documeFinal_Meeting_Minutes.xlsx	.xlsx	35840	1		FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive

Answer: Final_Annual_Report.xlsx

3. Which file was deleted? (for example: randomname.pdf)

Filtering for "InUse = False":

InUse	Parent1	Parent2	Parent3	FileName	Extension	FileSize	RefersTo	Repars	IsDirectory	HasAds	IsAds	Si<FN	uSecZe
FALSE				1.\documeMarketing_Plan.xlsx	.xlsx	45056	1		FALSE	FALSE	FALSE	FALSE	FALSE

Answer: Marketing_Plan.xlsx

4. How many of them have been set in Hidden mode? (for example: 43)

Filtering for "SiFlags = Hidden":

	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	ParentID	FileName	Extension	FileSize	Refere	Repars	IsDirectory	HasAds	IsAds	SI<FN	uSecZe	Copied	SiFlags	NameT	Create	Create
45	1	.\docume	credentials.txt	.txt	31	1	FALSE	FALSE	FALSE	Sort A to Z				Posix	32:27.3	
48										Sort Z to A						
49										Sort by Color						
50										Sheet View						
51										Clear Filter From "SiFlags"						
52										Filter by Color						
53										Text Filters						
54										Search						
55										☒ (Select All)						
56										☐ Archive						
57										☒ Hidden						
58										☐ Hidden[System						
59										☐ Hidden[System]IndexView						
60										OK	Cancel					
61																
62																
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Answer: 1

5. Which is the filename of the important TXT file that was created? (for example: randomname.txt)

Filtering for "Extension" column:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	EntryNi	Sequer	InUse	ParentID	ParentID	ParentID	FileName	Extension	FileSize	Refere	Repars	IsDirectory	HasAds	IsAds	SI<FN	uSecZe
45	51	1	TRUE	36	1	.\docume	credentials.txt	.txt	31	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
48																
49																
50																
51																
52																
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Answer: credentials.txt

6. A file was also copied, which is the new filename? (for example: randomname.pdf)

Filtering for "Copied = True" column:

	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Parent	FileName	Extension	FileSize	Refere	Repars	IsDirectory	HasAds	IsAds	Si<FN	uSecZe	Copied	SiFlags	NameT	Create	Create
46	1.	Financial_Statement_draft.xlsx	.xlsx	59392	1		FALSE	FALSE	Sort Smallest to Largest				Archive	Posix	32:27.3	
48									Sort Largest to Smallest							
49									Sort by Color							
50									Sheet View							
51									Clear Filter From "Copied"							
52									Filter by Color							
53									Number Filters							
54									Search							
55									☒ (Select All)							
56									☐ FALSE							
57									☒ TRUE							
58									OK							
59									Cancel							
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Answer: Financial_Statement_draft.xlsx

7. Which file was modified after creation? (for example: randomname.pdf)

Checking "Created0x10" and "LastModified0x10" and comparing them. If there is a difference it's the wanted file. By applying the following filter `=IF(TEXT(A2, "mm/dd/yyyy hh:mm:ss AM/PM") <> TEXT(B2, "mm/dd/yyyy hh:mm:ss AM/PM"), C2, "")` on each line we can quickly see which columns have different values.

AE2																
	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG
14	SE	Hidden	S DosWindc	32:21.7		32:21.7		32:21.7	32:21.7		0	1070159	256			
15	SE	Hidden	S DosWindc	32:21.7		32:21.7		32:21.7	32:21.7		0	1070159	256			
17	SE	Hidden	S Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1070176	257			
18	SE	Hidden	S Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1070210	257			
19	SE	Hidden	S Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1070193	257			
21	SE	Hidden	S Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1070227	257			
22	SE	Hidden	S Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1070227	257			
26	SE	Hidden	S Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1071438	258			
27	SE	Hidden	S Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1071438	258			
28	SE	Archive	Posix	32:22.3		32:22.3		32:22.3	32:22.3		0	1071835	259			
30	SE	Archive	Posix	32:24.3		32:24.3		32:24.3	32:24.3	32:24.3	0	1081036	262			
33	SE	Archive	Posix	32:27.3		32:27.3	32:27.3	32:27.3	32:27.3	32:27.3	0	1074516	264			
34	SE	Archive	Posix	32:27.3		32:27.3		32:27.3	32:27.3		0	1074860	264			
35	SE	Archive	Posix	32:27.3		32:27.3		32:27.3	32:27.3		0	1075184	264			
36	SE	Archive	Posix	32:27.3		32:27.3		32:27.3	32:27.3		0	1075516	264			
37	SE	Archive	Posix	32:27.3		32:27.3		32:27.3	32:27.3		0	1076129	264			
38	SE	Archive	Posix	32:27.3		32:27.3	32:27.3	32:27.3	32:27.3	32:27.3	0	1076478	264			
40	SE	Archive	Posix	32:27.3		32:27.3	32:27.3	32:27.3	32:27.3	32:27.3	0	1077300	264			
41	SE	Archive	Posix	2/20/2024 19:32		2/20/2024 19:33	32:27.3	33:30.3	32:27.3	33:30.3	32:27.3	0	1080575	264	Project_P	proposal.pdf
42	SE	Archive	Posix	2/20/2024 19:32		2/20/2024 19:32	32:27.3	32:27.3	32:27.3	32:27.3	0	1077973	264			
43	SE	Archive	Posix	32:27.3		32:27.3		32:27.3	32:27.3		0	1078631	264			
44	SE	Archive	Posix	32:27.3		32:27.3		32:27.3	32:27.3		0	1078959	264			
45	SE	Hidden	Posix	32:27.3		32:27.3		33:30.3	32:27.3	32:27.3	0	1080714	264			
46	JE	Archive	Posix	32:27.3		32:27.3	32:27.3	32:27.3	32:27.3	32:27.3	0	1079873	264			
47	SE	Archive	Posix	32:27.3		32:27.3		32:27.3	32:27.3		0	1080677	264			
48																
49																

Answer: Project_Proposal1.pdf

8. What is the name of the file located at record number 45? (for example: randomname.pdf)

Checking the "EntryNumber" column:

G40																	fx		Annual_Report.xlsx																
	A	B	C	D	E	F	G		H	I	J	K	L	M	N	O	P																		
1	EntryN	Sequer	InUse	Parent	Parent	Parent	FileName	Extension	FileSize	Refere	Repars	IsDirectory	HasAds	IsAds	SI<FN	uSecZe																			
40	45	1	TRUE	44	1	.\docume	Annual_Report.xlsx	.xlsx	61440	1		FALSE	FALSE	FALSE	FALSE	FALSE																			
48																																			
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Answer: Annual_Report.xlsx

9. What is the size of the file located at record number 40? (for example: 1337)

Checking the "EntryNumber" column:

135	57344															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	EntryN	Sequer	InUse	ParentI	ParentI	ParentI	FileName	Extension	FileSize	Refere	Repars	IsDirectory	HasAds	IsAds	SI<FN	uSecZe
35	40	1	TRUE	37	1	.\docume	Final_Project_Proposal.pdf	.pdf	57344	1		FALSE	FALSE	FALSE	FALSE	FALSE
48																
49																
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Answer: 57344